

Migration plan: common Morris-compatible Results sections across all nine pmsimstats manuscripts

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Objective

Bring all nine manuscripts in `analysis/report/01-` through `09-` into a common Results-section format that meets the Morris, White and Crowther (2019) ADEMP performance-measure recommendations end-to-end. The format must be reproducible (a single shared infrastructure package generates the tables and figures rather than each paper rolling its own), comparable (rows and columns are the same across papers, so cross-paper synthesis is trivial), and complete (every reported estimate is accompanied by its Monte Carlo standard error).

The plan accepts the cost of **rerunning simulations where necessary** to satisfy the format. Some papers' existing simulations preserved only summary outputs that lack the per-replicate granularity needed to compute MCSE retrospectively; those papers need their simulations rerun with the canonical output schema.

What 'common Morris-compatible' means

Canonical Results-section structure

Every paper's Results section follows the same six-subsection structure:

```
# Results

## Setup and replicate counts
(Per-cell n_reps, total cell count, total compute time,
 parameter-grid summary; confirms what was actually run.)

## Power and type I error
(Power for primary estimand at each cell, with binomial-
 proportion MCSE; type I error under each null cell, with
 same MCSE.)

## Bias of the primary estimand
(Mean bias and Monte Carlo SE of the bias; reported as
 absolute and as relative to the calibrated true effect.)

## Coverage of nominal-95% CIs
(Empirical coverage rate of nominal 95% CIs at each cell,
 with binomial-proportion MCSE; coverage of the model SE
 against the empirical SE across replicates.)

## Convergence and computational cost
(Convergence rate per cell, mean fit time, any cells with
 convergence below 90% flagged.)

## Sensitivity and cross-cell synthesis
(Paper-specific axis-by-axis sensitivity plots; any
 cross-paper comparisons that are made.)
```

Papers with multiple substudies (e.g., 02 with three carryover specs, 03 with five studies, 04 with M1 plus S1-S12, 09 with four studies) repeat the six-subsection block per substudy, with a synthesis subsection at the end.

Canonical results-table format

Every per-cell Results table has the same column structure:

cell	n_reps	power	mcse(power)	mcse(bias)	mcse(empSE)	mcse(empSE)	coverage	mcse(coverage)	coverage_rate
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where `cell` is a paper-specific identifier (e.g., `(N, t1/2, design)`), `n_reps` is the per-cell replicate count, and the remaining columns are the Morris ADEMP performance measures with their MCSEs.

Canonical figure conventions

- **Power-by-axis plots:** x-axis is the perturbed parameter, y-axis is power $\in [0, 1]$, error bars are 95% binomial CIs (Wilson interval), faceting by other axes.
- **Bias-by-axis plots:** x-axis is the parameter, y-axis is $\hat{\beta} - \beta_{\text{true}}$, error bars are $\pm 1.96 \cdot \text{MCSE}(\hat{\beta})$.
- **Heatmaps:** cells coloured by the primary performance measure (typically power); each cell annotated with the point estimate and its MCSE in subscript or in a tooltip.
- **Coverage plots:** x-axis is the parameter, y-axis is empirical coverage with a horizontal reference line at 0.95; error bars as binomial 95% CIs.
- **Palette:** viridis or cividis (colour-blind safe); identical across all nine papers.

Per-replicate output schema

Every simulation driver writes per-replicate output to `analysis/data/<paper-slug>/<study-slug>-replicates.rds` with the schema:

```
data.table(  
  cell_id, rep_idx,  
  beta_hat, beta_se, p_value, ci_lo, ci_hi, converged, fit_time  
)
```

plus paper-specific columns. The per-replicate granularity is mandatory because all Morris performance measures (especially MCSE of bias, empirical SE, and coverage) require recomputation from individual replicates rather than summary-only output. Any existing driver that writes only per-cell summaries must be modified to write per-replicate data.

Shared infrastructure: `tools/morris-helpers.R`

A single file at `tools/morris-helpers.R` provides the helper functions used by every paper's Results-generation chunks:

```
## morris_summary(replicates, by, beta_true)  
## Takes a data.table of per-replicate output and a vector of  
## grouping columns; returns a per-cell Morris-style summary  
## table with all performance measures and their MCSEs.  
  
morris_summary <- function(replicates, by, beta_true) { ... }  
  
## morris_table(summary_df)  
## Renders the summary as a kable table with the canonical  
## column structure above; suitable for inline use in an Rmd  
## via knitr::kable() with longtable / booktabs styling.  
  
morris_table <- function(summary_df, caption = NULL) { ... }  
  
## morris_power_plot(summary_df, x_axis, facet = NULL)
```

```

## Power-by-axis plot with binomial 95% CI error bars.

morris_power_plot <- function(summary_df, x_axis,
                             facet = NULL) { ... }

## morris_bias_plot(summary_df, x_axis, facet = NULL)
## Bias-by-axis plot with bias-MCSE error bars.

morris_bias_plot <- function(summary_df, x_axis,
                             facet = NULL) { ... }

## morris_heatmap(summary_df, x_axis, y_axis, fill = 'power')
## Heatmap of any performance measure across two axes; cells
## annotated with point estimates and MCSEs.

morris_heatmap <- function(summary_df, x_axis, y_axis,
                             fill = 'power') { ... }

```

These functions are sourced once at the top of each paper's `setup` chunk:

```
source(here::here('tools', 'morris-helpers.R'))
```

and used directly in inline R chunks throughout the Results section. Building this shared infrastructure is the first deliverable of the migration; once it exists, the per-paper work becomes mostly mechanical.

Per-paper migration assessment

Paper 01-dgp-mean-moderation-vs-mvn

Status: substantive simulation discussion exists, but uses LaTeX `\section{}` numbering rather than mark-down headings; no dedicated `# Results` section; existing power tables lack MCSE columns.

Existing data: simulation outputs are preserved at `analysis/data/figure4/` (per the `analysis/scripts/figure4/` driver). Per-replicate granularity present.

Migration action: 1. Reorganise the body sections so that the simulation results move from §3 and §6 into a dedicated `# Results` section following the canonical six-subsection structure. 2. Convert `\section{}` numbering to markdown `#/#` headings. 3. Recompute MCSE columns from preserved per-replicate output; no rerun needed. 4. Replace ad-hoc tables and figures with `morris_table()` and `morris*_plot()` calls.

Rerun needed? No — per-replicate data preserved.

Estimated effort: 1 day of writing; ~4 hours for figure regeneration and verification.

Paper 02-carryover-sensitivity

Status: substantive Results section already in place, with power, bias, type I error, coverage, and MCSE language throughout. The closest paper to canonical-format compliance.

Existing data: per-replicate output preserved at `analysis/data/02-grid-summary.rds` and `analysis/data/02-sensitivity-summary.rds`. Confirmed per-replicate granularity.

Migration action: minor reformatting to align column structure with the canonical `morris_table()` output. Replace existing custom tables and figures with the shared- infrastructure helpers.

Rerun needed? No — per-replicate data preserved.

Estimated effort: 0.5 days reformatting + 4 hours for table/figure regeneration.

Paper 03-latent-class-mixture-application

Status: conceptual/methods paper with no Results section yet. The simulation programme's Phase 1 results are in the Discussion's 'Progress to date' subsection. Pre-registered five-study programme committed in `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`.

Existing data: Phase 1 pilot results at `analysis/scripts/latent-class-mixture-application/output/03-study5-pilot.rds` (per-replicate). The pre-registered Studies 1-4 production runs have not been executed.

Migration action: 1. Run the pre-registered Studies 1-5 production simulations per the ADEMP pre-registration document. Total cell count ~2,300 (Study 1 alone is now 1,152 cells with the new `pb_class_correlation` axis). 2. Move the Phase 1 pilot results from the Discussion to a new `# Results` section under the canonical structure. 3. Populate the canonical Results subsections from the production-run output.

Rerun needed? Yes — **partial run already complete (Study 5 pilot, 100 reps); Studies 1-4 not yet run.**

Estimated compute: ~80 hours on 16 cores parallel for Studies 1-3; Study 4 ~8 hours; total programme ~90-100 core-days. Feasible over 2 weeks of background compute on a workstation.

Paper 04-treatment-main-effect

Status: nominal Results section exists but is a stub. ADEMP pre-registration committed in `analysis/scripts/treatment-main-effect/00-ademp-pre-registration.md` covering M1 (primary comparison) and S1-S12 (sensitivity sweeps).

Existing data: simulation infrastructure in `analysis/scripts/treatment-main-effect/`. Whether the production cells have been run requires checking `analysis/data/derived_data/sim_workspace.RData`.

Migration action: 1. Verify production-run output at `analysis/data/derived_data/sim_workspace.RData` is per-replicate (not just summary) and aligned with the pre-registration cell counts. 2. If yes: populate the canonical Results section from existing data. 3. If no: extend the driver to write per-replicate output and rerun M1 plus S1-S12.

Rerun needed? **Likely partial** — some sweeps may have summary-only output; rerun affected ones with the canonical schema.

Estimated compute: M1 alone is ~1 hour per cell $\times$ 8 cells = 8 core-hours; S1-S12 sweeps ~30 core-hours total; full programme ~40 core-hours, feasible overnight on 8 cores.

Paper 05-nof1-design-sensitivity

Status: substantive Results section in place, fully compliant on power and MCSE for the twelve sensitivity sweeps, but **coverage absent**. Sweeps S1-S12 reported with MCSE error bars but no coverage analysis.

Existing data: per-replicate output preserved at `analysis/data/05-sensitivity-replicates.rds` (verify).

Migration action: 1. Add a coverage subsection per sweep, computing nominal-95% CI coverage with binomial MCSE. 2. Reformat existing tables and figures to use the shared helpers.

Rerun needed? **Probably no** — if per-replicate output preserves CI bounds, coverage is computable retrospectively. If only point estimates and SEs were preserved, a rerun may be needed to record CI bounds. Verification step: check the rds schema.

Estimated effort: 1 day if no rerun needed; 2 weeks of background compute if S1-S12 must be rerun.

Paper 06-component-decomposition

Status: methods paper with worked examples; the §6 simulation-study placeholder describes the planned simulation but no Results section exists. Pre-registration committed in `analysis/scripts/component-`

decomposition/00-ademp-pre-registration.md covering three studies (A, B, C).

Existing data: none — simulation has not been executed.

Migration action: 1. Implement the three studies (Study A: bias under three-component DGP; Study B: identifiability of full decomposition; Study C: subadditivity sensitivity). 2. Run the production simulations per pre-registration. 3. Populate a new # Results section under the canonical structure.

Rerun needed? Yes — full programme not yet run.

Estimated compute: Study A is ~120 hours on 16 cores; Study B ~40 hours; Study C ~60 hours; total ~220 core-days. Largest single budget in the programme; needs dedicated compute.

Paper 07-gompertz-evaluation

Status: stub Results section. Pre-registration committed in analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.md covering Study 0 (calibration), Study 1 (power across families), and Study 2 (sloppiness diagnostic).

Existing data: none.

Migration action: 1. Run Study 0 (calibration sub-study). 2. Run Study 1 (4 families $\times$ 144 cells $\times$ 1000 reps). 3. Run Study 2 (sloppiness eigenvalue diagnostics). 4. Populate canonical Results section.

Rerun needed? Yes — full programme not yet run.

Estimated compute: ~50 core-days (Study 1 dominates).

Paper 08-test-procedure-design-sensitivity

Status: stub Results section. Pre-registration committed in analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-registration.md covering Study 1 (test comparison), Study 2 (cycle-by-period grid), Study 3 (joint design-by-test heatmap).

Existing data: none.

Migration action: 1. Implement the three test-procedure variants (RM-ANOVA, linear-mixed, GEE) and the cycle-by-period DGP grid in implementations/simple/simulation.R. 2. Run Studies 1-3 per pre-registration. 3. Populate canonical Results section.

Rerun needed? Yes — full programme not yet run.

Estimated compute: ~80 core-days (Study 2 has 240 cells).

Paper 09-informative-dropout-by-design

Status: stub Results section. Pre-registration committed in analysis/scripts/informative-dropout-by-design/00-ademp-pre-registration.md covering four studies (power $\times$ dropout, two-mode bias, ‘happy accident’ decomposition, dropout-pattern misspecification).

Existing data: none.

Migration action: 1. Extend pmsimstats R/censordata.R infrastructure with the explicit dropout-pattern grid. 2. Run Studies 1-4 per pre-registration. 3. Populate canonical Results section.

Rerun needed? Yes — full programme not yet run.

Estimated compute: ~30 core-days (~370 cells total across the four studies).

Summary of compute budget

Paper	Per-replicate data preserved?	Rerun?	Compute estimate
01	yes	no	format only
02	yes	no	format only
03	partial (Study 5 pilot)	yes (Studies 1-4)	~90 core-days
04	likely partial	likely partial	~40 core-hours
05	needs verification	possibly no	format if no; 2 weeks if yes
06	no	yes (full)	~220 core-days
07	no	yes (full)	~50 core-days
08	no	yes (full)	~80 core-days
09	no	yes (full)	~30 core-days

Total compute: roughly 500-600 core-days at the upper bound. On a 16-core workstation running 24/7, this is approximately 4-5 weeks of dedicated background compute. On a smaller machine running 8 cores in non-blocking background, 2-3 months. Feasible at the project’s typical research-programme timescale.

Sequencing

The migration is staged in five phases. Each phase is a discrete deliverable; a paper that has reached the end of its phase is fully canonical-format compliant.

Phase 1: Build shared infrastructure (1-2 weeks)

- Implement `tools/morris-helpers.R` with the six helper functions (`morris_summary`, `morris_table`, `morris_power_plot`, `morris_bias_plot`, `morris_heatmap`, plus a setup helper that loads `kableExtra`, `ggplot2`, etc.).
- Document the per-replicate output schema in `analysis/data/README-replicate-schema.md`.
- Commit unit tests of the helpers against synthetic data.
- Verify the helpers can read existing per-replicate outputs from papers 01 and 02 (to confirm the schema is realistic).

Deliverable: a single source-able R file that any paper can use; tested helpers; documented schema.

Phase 2: Reformat papers 01, 02, 05 (1-2 weeks)

The three papers with substantive existing simulation outputs get their Results sections reformatted using the Phase 1 helpers. No reruns required for papers 01 and 02. Paper 05 may need a rerun for coverage; resolved by verification before committing to a rerun.

Deliverable: papers 01, 02, 05 with canonical-format Results sections, populated from existing data.

Phase 3: Populate paper 04 (1 week + overnight compute)

Verify paper 04’s existing simulation output schema. Populate the canonical Results section from per-replicate data; rerun any sweeps that lack per-replicate granularity.

Deliverable: paper 04 with canonical-format Results section.

Phase 4: Run and populate stub papers 07, 08, 09 (3-4 weeks)

Build the simulation drivers for papers 07, 08, 09. Run the production cells per pre-registration. Populate canonical Results sections.

Order: 09 (smallest compute) → 07 → 08 (largest compute). Each can run as background while the next is being prepared.

Deliverable: papers 07, 08, 09 with canonical-format Results sections.

Phase 5: Run papers 03 and 06 (4-6 weeks)

Largest compute commitment of the programme. Execute the five-study latent-class programme (03) and the three-study component-decomposition programme (06).

These run in parallel during the writing phase; the Results sections are populated as compute completes per study.

Deliverable: papers 03 and 06 with canonical-format Results sections.

Cross-cutting requirements

- **Pre-registration discipline.** Each Results section is populated from cells specified in the corresponding ADEMP pre-registration document. Any deviation goes into `02-deviations-log.md` for that paper before the production runs commit.
- **Per-replicate output mandatory.** Every driver writes per-replicate data to `analysis/data/<paper-slug>/. No driver writes summary-only output as its primary checkpoint (summaries can be derived from per-replicate; the converse is impossible).`
- **Reproducibility.** Each paper's Results section is generated from its per-replicate rds via the shared helpers on render. The Results section text is mostly fixed prose with inline R chunks calling the helpers; rerendering after a fresh simulation run regenerates the tables and figures automatically.
- **Cross-paper synthesis.** A planned Phase-6 cross-paper discussion document at `analysis/report/cross-paper-synthesis.md` will use the shared helpers' canonical output to compare results across papers without per-paper formatting differences obstructing the comparison.

Risks and mitigations

Risk	Mitigation
Some papers' existing per-replicate data are missing or in non-canonical form	Verification step at the start of each paper's phase; rerun if necessary
Compute budget exceeds available capacity	Run papers in priority order; defer Phase 5 until Phases 2-4 complete
Helper functions need iteration once real papers exercise them	Phase 1 includes unit tests; helper API may evolve in early Phase 2 papers
ADEMP pre-registration parameters need revision after Phase 2 results inform expectations	Deviation log mechanism already in place per paper
Coverage analysis on rerun-resistant papers blocked	Document the gap and run dedicated short coverage sweeps where the full rerun is infeasible

Deliverables (consolidated)

By the end of the migration:

- `tools/morris-helpers.R` and unit tests in `tools/tests/test-morris-helpers.R`.
- `analysis/data/README-replicate-schema.md`.
- Each paper's Results section in canonical-format markdown, populated from per-replicate output via the shared helpers.
- Each paper's per-replicate output preserved at `analysis/data/<paper-slug>/<study-slug>-replicates.rds`.
- Each paper's ADEMP pre-registration committed (already in place after the previous turn) with `02-deviations-log.md` recording any in-flight changes.
- A cross-paper synthesis document at `analysis/report/cross-paper-synthesis.md` (Phase 6, optional).

Source: `~/Dropbox/prj/alz/10-pmsimstats-ng/pmsimstats-ng/analysis/report/morris-results-migration-plan.md`